

RSA and Period Finding:  
Foundations, Resonance Formulation, Limits

## Contents

|     |                                                     |   |
|-----|-----------------------------------------------------|---|
| .1  | Introduction to RSA Cryptography                    | 3 |
|     | Mathematical Foundation                             | 3 |
|     | Key Generation                                      | 3 |
|     | Encryption and Decryption                           | 3 |
| .2  | Mathematical Proofs                                 | 3 |
|     | Correctness Proof                                   | 3 |
| .3  | Implementation                                      | 3 |
|     | Modular Exponentiation                              | 3 |
|     | Prime Generation                                    | 4 |
| .4  | Security Analysis                                   | 4 |
|     | Common Attacks                                      | 4 |
|     | Recommendations                                     | 4 |
| .5  | Performance Analysis                                | 4 |
|     | Optimisations                                       | 4 |
| .6  | Extensions                                          | 5 |
|     | Multi-Prime RSA                                     | 5 |
|     | Blinding                                            | 5 |
| .7  | The arithmetic foundation                           | 7 |
| .8  | The spectral formulation                            | 7 |
|     | Why period finding is a resonance problem           | 7 |
|     | Primes as generators                                | 7 |
|     | What the grid parameter $\sigma$ achieves           | 7 |
| .9  | Topology: winding number instead of frequency ratio | 7 |
| .10 | Structural scaling limit                            | 7 |
|     | The Weyl obstruction                                | 7 |
|     | The precision barrier                               | 7 |
| .11 | Comparison of methods                               | 8 |
| .12 | What the Fourier transform achieves                 | 8 |
| .13 | Balance                                             | 8 |
| .14 | Verification                                        | 8 |
| .15 | Relation to the corpus                              | 8 |

## Summary

This document treats the RSA procedure in two parts. Part I presents the number-theoretic foundations and security analysis. Part II examines the attack surface via period finding as a resonance problem and determines its carriers and limits: the resonance description is correct – period finding is a spectral problem – but its generators are the **primes**, not an adjustable parameter.

**Two separate parameters**

This document uses two quantities that must not be confused:

- $\xi = 4/30000$  – the FFGFT geometry parameter. Fixed, not adjustable, with no operational connection to factorisation.
- $\sigma$  – a numerical grid parameter of the search algorithms (values such as  $1/42$ ,  $1/100$ ,  $1/1000$ ). A sampling quantity, not a physical one.

## **Part I: The RSA Procedure**

## .1 Introduction to RSA Cryptography

The RSA algorithm (Rivest, Shamir, Adleman 1977) is one of the first practical public-key cryptosystems and is widely used for secure data transmission.

### Mathematical Foundation

RSA is based on the computational difficulty of factoring large integers and the properties of modular arithmetic.

### Key Generation

1. Choose two distinct prime numbers  $p$  and  $q$ .
2. Compute  $n = p \cdot q$ .
3. Compute Euler's totient:  $\varphi(n) = (p - 1)(q - 1)$ .
4. Choose  $e$  with  $1 < e < \varphi(n)$  and  $\gcd(e, \varphi(n)) = 1$ .
5. Compute  $d$  with  $d \cdot e \equiv 1 \pmod{\varphi(n)}$ .

The public key is  $(n, e)$ , the private key  $(n, d)$ .

### Encryption and Decryption

For a message  $M$  with  $0 \leq M < n$ :

$$C \equiv M^e \pmod{n} \quad (\text{encryption}), \quad (1)$$

$$M \equiv C^d \pmod{n} \quad (\text{decryption}). \quad (2)$$

## .2 Mathematical Proofs

### Correctness Proof

**Theorem .2.1** (RSA Correctness). *For any message  $M$  with  $0 \leq M < n$  and  $\gcd(M, n) = 1$ :  $(M^e)^d \equiv M \pmod{n}$ .*

*Proof.* From  $ed \equiv 1 \pmod{\varphi(n)}$  follows  $ed = 1 + k\varphi(n)$ . By Euler's theorem  $M^{\varphi(n)} \equiv 1 \pmod{n}$ , so

$$C^d \equiv M^{ed} \equiv M^{1+k\varphi(n)} \equiv M \cdot (M^{\varphi(n)})^k \equiv M \pmod{n}.$$

For  $\gcd(M, n) \neq 1$  the Chinese Remainder Theorem gives the same. □

## .3 Implementation

### Modular Exponentiation

Efficient modular exponentiation by square-and-multiply:

**Algorithm: Modular Exponentiation****Function** ModExp(*base*, *exponent*, *modulus*):

1.  $result \leftarrow 1$ ;  $base \leftarrow base \bmod modulus$
2. **while**  $exponent > 0$ :
  - (a) **if**  $exponent \bmod 2 = 1$ :  $result \leftarrow (result \times base) \bmod modulus$
  - (b)  $exponent \leftarrow \lfloor exponent/2 \rfloor$ ;  $base \leftarrow (base \times base) \bmod modulus$
3. **return**  $result$

**Prime Generation**

- Probabilistic primality tests (Miller-Rabin).
- $p$  and  $q$  of similar bit length.
- Avoid special forms that are easier to factor.

**.4 Security Analysis****Common Attacks**

| Attack         | Description                         |
|----------------|-------------------------------------|
| Factorisation  | Factor $n$ into $p$ and $q$         |
| Small $e$      | Certain messages recoverable        |
| Timing attacks | Deduce secret from computation time |
| Side-channel   | Power, electromagnetic leaks        |

**Recommendations**

1. Key sizes  $\geq 2048$  bit (3072 or 4096 for long-term security).
2. Padding scheme OAEP.
3. Constant-time implementations.
4. Regular updates to cryptographic libraries.

**.5 Performance Analysis**

| Operation      | Complexity | Typical time (2048 bit) |
|----------------|------------|-------------------------|
| Key generation | $O(k^3)$   | 1–10 seconds            |
| Encryption     | $O(k^2)$   | < 1 ms                  |
| Decryption     | $O(k^3)$   | 10–100 ms               |

**Optimisations**

- Chinese Remainder Theorem for faster decryption.
- Windowing methods for exponentiation.
- Hardware acceleration.

## .6 Extensions

### Multi-Prime RSA

$n = p_1 p_2 \cdots p_k$ : faster decryption via multi-prime CRT.

### Blinding

Protection against timing attacks:

$$C' = C \cdot r^e \pmod{n}, \quad M' = (C')^d \pmod{n}, \quad M = M' \cdot r^{-1} \pmod{n}.$$

## **Part II: Period Finding as a Resonance Problem**

## .7 The arithmetic foundation

Factorisation of  $n = pq$  reduces to period search. For random  $a$  with  $\gcd(a, n) = 1$  the period  $r$  satisfies  $a^r \equiv 1 \pmod{n}$ . If  $r$  is even and  $a^{r/2} \not\equiv -1 \pmod{n}$  then  $\gcd(a^{r/2} \pm 1, n)$  yields the factors.

**[K]** The period  $r$  always divides  $\lambda(n) = \text{lcm}(p-1, q-1)$ . It is an arithmetic quantity, not determined by a choosable parameter.

## .8 The spectral formulation

### Why period finding is a resonance problem

The sequence  $f(x) = a^x \bmod n$  is periodic with period  $r$ . Its Fourier transform has sharp maxima at  $k/r$ . Period search is formally spectral analysis.

### Primes as generators

**[K]** The resonance structure rests on primes – measured by the bicoherence test on 2469 Riemann zeros (Doc. 316, August 2026): coupling at prime powers ( $b^2 = 0.76\text{--}0.85$ ), none at composites ( $b^2 = 0.018\text{--}0.025$ ), contrast factor 30–50.

### What the grid parameter $\sigma$ achieves

**[X]** All  $\sigma$  values have composite denominators and form mixed points without their own resonance line. Their function is that of a sampling grid.

## .9 Topology: winding number instead of frequency ratio

**[K]** For closed trajectories the winding number  $W = \log_2(r)$  decides, not  $r$ . Pure fifth:  $W = 0.5850$  (irrational)  $\rightarrow$  open. Tempered:  $W = 7/12$  (rational)  $\rightarrow$  closed after 12 steps. Closure is an artefact of rationalisation.

## .10 Structural scaling limit

### The Weyl obstruction

**[K]** The Laplace spectrum of every compact manifold grows as  $N(\lambda) \sim C_d \lambda^{d/2}$ ; the arithmetic counting density as  $N(T) \sim T \ln T$  – not a power. A compact resonance space cannot carry a spectrum of density  $T \ln T$ .

**Delimitation.** This obstruction does not apply to Shor's algorithm. Shor seeks one period of one concrete number (finite task,  $2n+3$  qubits). What limits Shor at large  $n$  is the gate count ( $\sim n^3$ ) and error-correction overhead – a *resource limit*.

### The precision barrier

For sharp separation at  $r \approx n$  a resolution  $\Delta f \approx f_0/n^2$  is needed – at RSA sizes below any available number representation.

## .11 Comparison of methods

| Method           | Principle            | Closure             | Limit          |
|------------------|----------------------|---------------------|----------------|
| Trial division   | Division             | Discrete arithmetic | $O(\sqrt{n})$  |
| Pollard $\rho$   | Cycle detection      | Birthday paradox    | $O(n^{1/4})$   |
| Resonance filter | Spectral search      | Rationalised grid   | $O(n)$         |
| Shor's QFT       | Quantum interference | Discrete phase      | resource limit |

Preparation (modular exponentiation) accounts for approximately **95 %** of the gate cost; the QFT for only 5 % (Doc. 176, August 2026).

**[X]** A general exponential quantum advantage is not established.

## .12 What the Fourier transform achieves

**[K]** The Fourier transform is a *projection*: it translates a representation without generating new information. Measured on the Riemann zeros (Doc. 316): peaks at  $\ln 2, \ln 3, \ln 5, \ln 7, \ln 11, \ln 13$  where information is present; no peaks at composite arguments.

## .13 Balance

| Statement                                    | Status                             |
|----------------------------------------------|------------------------------------|
| RSA correctness (Euler, CRT)                 | <b>[K]</b> proved                  |
| Period finding is a spectral problem         | <b>[K]</b>                         |
| Primes are the generators                    | <b>[K]</b> measured by bicoherence |
| Winding number decides closure               | <b>[K]</b>                         |
| Preparation $\approx 95\%$ of gate cost      | <b>[K]</b>                         |
| $\sigma$ grids generate resonances           | <b>[X]</b> sampling artefact       |
| $\xi = 4/30000$ contributes to factorisation | <b>[X]</b> geometry                |
| Classical period search $O((\log n)^3)$      | <b>[X]</b> refuted                 |
| Weyl obstruction limits Shor                 | <b>[X]</b> resource limit          |
| Fourier transform generates information      | <b>[X]</b> projection              |

## .14 Verification

Scripts in 2/python/Shor\_Skripte/ (standard library, assertions, seed 20780458):  
s1\_periodensuche.py, s2\_grenzen.py, s3\_thermik\_zustandsraum.py.

## .15 Relation to the corpus

Doc. 316 (August 2026): Weyl obstruction, bicoherence test, winding number, Fourier transform as projection. Revision declared under R75 in Doc. 190.